

习题 5

5-6. (1) $\dot{i}_L(0_+) = I_3$

$$I_1 = \frac{10}{2000 + \frac{4000}{2}} = 2.5 \times 10^{-3} \text{ A}$$

$$I_3 = I_1 \times \frac{1}{2} = 1.25 \times 10^{-3} \text{ A} = \dot{i}_L(0_+)$$

$$\dot{i}_L(\infty) = 0$$

$$\tau = \frac{L}{R} = \frac{0.2}{8000} = \frac{1}{40000} \text{ s}$$

$$\begin{aligned} i_L(t) &= \dot{i}_L(\infty) + (\dot{i}_L(0) - \dot{i}_L(\infty))e^{-40000t} \\ &= 1.25e^{-40000t} \times 10^{-3} \text{ A} \end{aligned}$$

$$\begin{aligned} (2) \quad u_L(t) &= -(R_2 + R_3) \dot{i}_L(t) \\ &= -10 e^{-40000t} \text{ V} \end{aligned}$$

5-8. (1) $\dot{i}_L(0_+) = 0$

$$\dot{i}_L(\infty) = \frac{200}{12 + \frac{12 \times 4}{16}} \times \frac{12}{12+4} = 10 \text{ A}$$

$$\tau = \frac{L}{R} = \frac{20 \times 10^{-3}}{10} = 2 \times 10^{-3} \text{ s}$$

$$\begin{aligned} i_L(t) &= 10 + (0 - 10)e^{-500t} \\ &= 10(1 - e^{-500t}) \text{ A} \end{aligned}$$

$$(2) \quad \dot{i}_L(3 \times 10^{-3}) = 10(1 - e^{-1.5}) \text{ A}$$

5-9.

$$U_C(0+) = 30 \times \frac{50}{250} = 6 \text{ V}$$

$$U_C(\infty) = 30 \text{ V}$$

$$\tau = RC = 200 \times 10 \times 10^{-6} = 2 \times 10^{-3} \text{ s}$$

$$U_C(t) = 30 + (6 - 30)e^{-500t}$$

$$= 30 - 24e^{-500t}$$

$$\dot{I}_C(t) = \frac{30 - U_C(t)}{200} = \frac{24e^{-500t}}{200} \text{ A}$$

5-10

$$c1) \quad \dot{I}_L(0-) = \frac{100 \text{ V}}{100 \Omega} = 1 \text{ A}$$

$$U_L(0-) = 0 \text{ V}$$

$$(2) \quad \dot{I}_L(0+) = \dot{I}_L(0-) = 1 \text{ A}$$

$$\dot{I}_L(\infty) = \frac{100}{100} + \frac{50}{50} = 2 \text{ A} \quad \checkmark \text{ 叠加原理}$$

$$\tau = \frac{L}{R} = \frac{5}{\frac{100 \times 50}{100 + 50}} = \frac{3}{20} \text{ s}$$

$$\dot{I}_L(t) = 2 + (1 - 2)e^{-\frac{20t}{3}} = 2 - e^{-\frac{20t}{3}} \text{ A}$$

$$U_L(t) = L \frac{d\dot{I}_L(t)}{dt} = \frac{100}{3} \times e^{-\frac{20t}{3}} \text{ V}$$

5-11

$$i_L(0+) = 4 \times \frac{2}{2 + \frac{4 \times 4}{4+4}} = 2 \text{ A}$$

$$U_C(0+) = 2 \times 2 = 4 \text{ V}$$

$$i_L(\infty) = 4 \times \frac{2}{2+4} = \frac{4}{3} \text{ A}$$

$$U_C(\infty) = 0$$

$$\tau_L = \frac{L}{R} = \frac{1}{2+4} = \frac{1}{6} \text{ s}$$

$$\tau_C = RC = 4 \times 1 = 4 \text{ s}$$

$$U_C(t) = 0 + (4-0) e^{-\frac{t}{4}} \text{ V}$$

$$= 4e^{-\frac{t}{4}} \text{ V}$$



$$i_L(t) = \frac{4}{3} + (2 - \frac{4}{3}) e^{-6t} \text{ A}$$

$$= \frac{4}{3} + \frac{2}{3} e^{-6t} \text{ A}$$

$$U_s = U - U_C(t) = \frac{16}{3} + \frac{8}{3} e^{-6t} - 4e^{-\frac{t}{4}} \text{ V}$$

5-1

$$U_C(0+) = U_C(0-) = 6 \text{ V}$$

$$I_3 = -\frac{6}{2.4 + 0.6} = -2 \text{ A}$$

$$I_2 = \frac{6 - 2 \times 0.6}{6} = 0.8 \text{ A}$$

$$I_1 = I_3 + I_2 = -1.2 \text{ A}$$

5-2

$$I_L(0+) = I_L(0-) = \frac{3V}{10+20} = 0.1A$$

$$I_1(0+) = 0.1A$$

$$I_2(0+) = 0.1 \times \frac{5}{20+5} = 0.02A$$

$$I_3(0+) = 0.1 \times \frac{20}{20+5} = 0.08A$$

5-3

$$u_C(0+) = u_C(0-) = 90 \times \frac{3}{3+1.5} = 60V$$

$$i_C(0-) = 0A$$

$$i_C(0+) = -\frac{60}{100000} = -6 \times 10^{-4}A$$

$$(2) \quad u_C(0-) = 0V$$

$$i_L(0+) = i_L(0-) = 0$$

$$u_C(0+) = -u_R = -90 \times \frac{2}{3} = -60V$$

$$(3) \quad u_R(0-) = u_R(0+) = 60V$$

$$i_R(0+) = i_R(0-) = \frac{90}{3000+1500} = 2 \times 10^{-2}A$$

5-4

$$u_C(0+) = u_C(0-) = 0$$

$$u_C(\infty) = 10 \times \frac{3}{2+3} = 6V$$

$$\tau = 100 \times 10^{-6} \times \frac{2 \times 3}{2+3} \times 10^3 = 0.12 \text{ s}$$

$$u_C(t) = 6(1 - e^{-\frac{25}{3}t}) \text{ V}$$

由 KVL 定律

$$u_2 + u_C(t) = 0$$

$$u_2 = 6(e^{-\frac{25}{3}t} - 1) \text{ V}$$

$$i_2(t) = \frac{u_2}{R_2} = 2(e^{-\frac{25}{3}t} - 1) \times 10^{-3} \text{ A}$$

5-5

$$u_C(0+) = u_C(0-) = 20 \text{ V}$$

$$\tau = RC = 2 \times 10^3 \times 50 \times 10^{-6} = 0.1 \text{ s}$$

$$u_C(t) = 20 e^{-10t} \text{ V}$$

$$i(t) = -\frac{20 e^{-10t}}{2 \times 10^3} = -e^{-10t} \times 10^{-2} \text{ A}$$

$$\therefore t = 200 \text{ ms 时}$$

$$u_C = 20 e^{-2} \text{ V}$$

0.2

$$i = -e^{-2} \times 10^{-2} \text{ A}$$

5-7

$$u_C(0+) = u_C(0-) = 0$$

$$u_C(\infty) = 30 \times \frac{1}{2} = 15 \text{ V}$$

$$\tau = 2 \times 10^{-6} \times 10 \times 10^3 = \frac{1}{50} \text{ s}$$

$$u_C(t) = 15(1 - e^{-50t}) \text{ V}$$

$\frac{1}{100}$
0.02

2×10^{-2}

$$\dot{i}_c(t) = C \frac{dU_c(t)}{dt} = 2 \times 10^{-6} = 1.5 \times 10^{-3} e^{-50t} \text{ A}$$

$$\begin{aligned} U_2 = U_3 + U_c &= 5 \times 10^3 \times 1.5 \times 10^{-3} e^{-50t} + 15(1 - e^{-50t}) \\ &= 15 - 7.5 e^{-50t} \end{aligned}$$

$$\dot{i}_2 = \frac{U_2}{R_2} = (1.5 - 0.75 e^{-50t}) \times 10^{-3} \text{ A}$$

$$\dot{i}_1 - \dot{i}_c(t) - \dot{i}_2 = 0$$

$$\dot{i}_1 = \dot{i}_2 + \dot{i}_c(t)$$

$$= (1.5 - 0.75 e^{-50t}) \times 10^{-3} + 1.5 \times 10^{-3} e^{-50t}$$

$$= (1.5 + 0.75 e^{-50t}) \times 10^{-3} \text{ A}$$